

MAC-106 Assignment 5

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Q1. Lossless Decomposition

We need to check if $(R_1 \cap R_2) \rightarrow R_1$ or $(R_1 \cap R_2) \rightarrow R_2$ holds.

Intersection: $R_1 \cap R_2 = \{A, B, C\} \cap \{A, D, E\} = \{A\}$

Now find A^+ using $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$:

- Start with $A^+ = \{A\}$
- From $A \rightarrow BC$, we get $\{A, B, C\}$

So $A^+ = \{A, B, C\} = R_1$.

Hence, $(R_1 \cap R_2) \rightarrow R_1$ holds, so decomposition is lossless.

Q2. Closures, Keys, and FDs

(a) Closure of A

- $A^+ = \{A\}$
- $A \rightarrow BC \Rightarrow \{A, B, C\}$
- $B \rightarrow D \Rightarrow \{A, B, C, D\}$
- $CD \rightarrow E \Rightarrow \{A, B, C, D, E\}$

So, $A^+ = \{A, B, C, D, E\}$.

(b) Candidate Keys

- A : gives all attributes, so it's a key
- E : since $E \rightarrow A$, and A gives everything, so E is also a key
- BC : using steps, it gives all attributes, so it's also a key
- CD : also gives everything

So candidate keys are: **A, E, BC, CD**

(c) Type of FDs

All given FDs are non-trivial since RHS is not a subset of LHS.

Q3. Functional Dependencies from Table

From observation:

- $A \rightarrow B$ holds
- $C \rightarrow B$ holds
- Others fail because same value maps to multiple outputs

So the main non-trivial FDs are: $A \rightarrow B, C \rightarrow B$

Q4. Canonical Cover

Start with:

$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Split RHS:

$F' = \{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

No extraneous attributes found.

Also, none of the FDs seem redundant after checking.

So final canonical cover is same as original:

$$F^c = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$$

Q5. Dependency Preservation

Given decomposition:

$R_1 = (A, B, C) \ R_2 = (A, D, E)$

Projected FDs:

- In R_1 : $A \rightarrow BC$
- In R_2 : $E \rightarrow A$

FDs like $B \rightarrow D$ and $CD \rightarrow E$ are lost.

So decomposition is **not dependency-preserving**.

Q6. Problems in Design

Redundancy: Same data repeats multiple times.

Example: department location repeated for many employees.

Insertion anomaly: Cannot insert some data unless other data exists.

Example: can't add department without an employee.

Q7. Normalization

(a) B^+

- $B \rightarrow D$
- $D \rightarrow A$
- $A \rightarrow BCD$
- $BC \rightarrow DE$

So $B^+ = \{A, B, C, D, E\}$

(b) AG is superkey

From $A \rightarrow BCDE$, adding G gives all attributes.
So AG is a superkey.

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(c) Canonical Cover

After simplification:

$$F^c = \{A \rightarrow BC, B \rightarrow DE, D \rightarrow A\}$$

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(d) 3NF Decomposition

Relations:

- (A, B, C)
- (B, D, E)
- (A, D)
- (A, G)

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(e) BCNF

Final decomposition:

$$(A, B, C, D, E), (A, G)$$

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Q8. BCNF Decomposition

Using $B \rightarrow D$:

- $R_1 = (B, D)$
- $R_2 = (A, B, C, E)$

Both are in BCNF.

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Q9. Why not BCNF always?

- May lose dependencies
- More joins required $\rightarrow$ slower queries

So sometimes 3NF is preferred.

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Q10. 4NF

4NF removes redundancy due to multivalued dependencies.

Example:

Split into:

- BookAuthors
- BookInfo
- TitlePub
- PhysicalBooks

Similarly for Users:

- Departments
- Users